Crafting the Ultimate Docker Image for Spring Applications

bellsoft

`whoami`

Pasha Finkelshteyn @asm0dey.site @asm0di0

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- And Spring
- Twitter asm0di0

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- And
- asm0di0
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BellSoft

- Vendor of Liberica JDK
- Contributor to the OpenJDK
- Author of ARM32 support in JDK
- Own base images
- Own Linux: Alpaquita

Liberica is the JDK officially recommended by

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We know our stuff!

bell/soft

bitly

So, what is ultimate?

- Smallest
- Fatsest startup
- Something inbetween

So, let’s look at all of them!

How do we create an image?

Let’s start from trivial.

```
FROM bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl

COPY . /app
RUN cd /app && ./gradlew build
ENTRYPOINT     java -jar /app/build/libs/spring-petclinic*.jar
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`Dockerfile` directives

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2. Layers are *immutable*
3. Some layers are zero-sized

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```
FROM bash

COPY . /app
RUN rm -rf /app
```

4. We would like image to be light, but it’s not :(

Our `Dockerfile`

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RUN cd /app && ./gradlew build
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```

Result

```
<missing>                                                 0B        ARG BUNDLE_TYPE
<missing>                                                 0B        ARG BUNDLE_TYPE
<missing>                                                 0B        ARG BUNDLE_TYPE
<missing>                                                 0B        ARG JAVA_RELEAS
<missing>                                                 7.49MiB   CMD ["/bin/sh"]
<missing>                                                 0B        |9 APK_REPOS= D
<missing>                                                 0B        ADD file:a4d77f
<missing>                                                 0B        LABEL org.openc
<missing>                                                 0B        LABEL org.openc
<missing>                                                 0B        LABEL maintaine
<missing>                                                 0B        ARG APK_REPOS D
<missing>                                                 0B        ARG APK_REPOS L
<missing>                                                 0B        ARG APK_REPOS L
<missing>                                                 0B        ARG APK_REPOS L
<missing>                                                 0B        ARG LIBC MINIRO
<missing>                                                 0B        ARG MINIROOTFS
```

Result

```
<missing>    0B        LABEL org.openc
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<missing>    0B        ARG BUNDLE_TYPE
<missing>    0B        ARG BUNDLE_TYPE
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Result

```
74a0788411                                                   512.03MiB /bin/sh -c cd /
7b82eb08ec                                                   67.41MiB  COPY dir:ef251b
e4bcf83d29 bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl 98.32MiB  |15 BUNDLE_TYPE
<missing>                                                    0B        ENV JAVA_HOME="
<missing>                                                    0B        ENV LANG=en_US.
<missing>                                                    0B        LABEL org.openc
<missing>                                                    0B        LABEL org.openc
<missing>                                                    0B        LABEL maintaine
<missing>                                                    0B        ARG BUNDLE_TYPE
<missing>                                                    0B        ARG BUNDLE_TYPE
<missing>                                                    0B        ARG BUNDLE_TYPE
<missing>                                                    0B        ARG BUNDLE_TYPE
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Result

```
ID         TAG                                                SIZE      COMMAND
cda5201e70 dumb:latest                                        0B        CMD ["/bin/sh"
74a0788411                                                    512.03MiB /bin/sh -c cd /
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<missing>                                                     0B        ARG BUNDLE_TYPE
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<missing>                                                      0B        ARG BUNDLE_TYPE
<missing>                                                      0B        ARG BUNDLE_TYPE
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<missing>                                                     0B        ARG BUNDLE_TYPE
<missing>                                                     0B        ARG BUNDLE_TYPE
<missing>                                                     0B        ARG BUNDLE_TYPE
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<missing>                                                       0B        ARG BUNDLE_TYPE
<missing>                                                       0B        ARG BUNDLE_TYPE
<missing>                                                       0B        ARG BUNDLE_TYPE
<missing>                                                       0B        ARG BUNDLE_TYPE
<missing>                                                       0B        ARG BUNDLE_TYPE
<missing>                                                       0B        ARG JAVA_RELEAS
```

579.44 MB are changed on every build!

Why do we care? We care because:

1. `push` takes longer time to start (update is longer)
2. `pull` takes longer (update takes longer & scaling takes longer)

Also, more disk psace is inefficiently used

Optimizing. Round 1.

Let's build it outside of container

```
FROM bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl

COPY . /app
RUN cd /app && ./gradlew build
ENTRYPOINT java -jar /app/build/libs/spring-petclinic*.jar
```

Optimizing. Round 1.

Let’s build it outside of container

```
FROM bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl

COPY build/libs/spring-petclinic-3.3.0.jar /app/app.jar
CMD java -jar /app/app.jar
```

Optimizing. Round 1.

Let’s build it outside of container

```
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COPY build/libs/spring-petclinic-3.3.0.jar /app/app.jar
CMD java -jar /app/app.jar
```

Just copying the prebuilt `jar` file

Results

```
ID         TAG                                                SIZE      COMMAND
cda5201e70 dumb:latest                                        0B        CMD ["/bin/sh" "-c" "/app
74a0788411                                                    512.03MiB /bin/sh -c cd /app && ./g
7b82eb08ec                                                    67.41MiB  COPY dir:ef251bd1e17ac0af
e4bcf83d29 bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl 98.32MiB |15 BUNDLE_TYPE=jdk CDS=n
<missing>                                                     0B        ENV JAVA_HOME="/usr/lib/j
<missing>                                                     0B        ENV LANG=en_US.UTF-8 LANG
<missing>                                                     0B        LABEL org.opencontainers.
<missing>                                                     0B        LABEL org.opencontainers.
<missing>                                                     0B        LABEL maintainer="$MAINTA
<missing>                                                     0B        ARG BUNDLE_TYPE CDS DESCR
<missing>                                                     0B        ARG BUNDLE_TYPE CDS JAVA_
<missing>                                                     0B        ARG BUNDLE_TYPE CDS JAVA_
<missing>                                                     0B        ARG BUNDLE_TYPE CDS JAVA_
<missing>                                                     0B        ARG BUNDLE_TYPE JAVA_RELE
<missing>                                                     0B        ARG JAVA_RELEASE
<missing>                                                     7.49MiB   CMD ["/bin/sh"]
```

Results

```
ID         TAG                                               SIZE     COMMAND
cefb30e21d smarter:latest                                    0B       CMD ["/bin/sh" "-c" "java
46d7594000                                                   58.80MiB COPY file:8d96bee95ae5ce43
e4bcf83d29 bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl 98.32MiB |15 BUNDLE_TYPE=jdk CDS=no
<missing>                                                    0B       ENV JAVA_HOME="/usr/lib/jv
<missing>                                                    0B       ENV LANG=en_US.UTF-8 LANGU
<missing>                                                    0B       LABEL org.opencontainers.i
<missing>                                                    0B       LABEL org.opencontainers.i
<missing>                                                    0B       LABEL maintainer="$MAINTAI
<missing>                                                    0B       ARG BUNDLE_TYPE CDS DESCRI
<missing>                                                    0B       ARG BUNDLE_TYPE CDS JAVA_R
<missing>                                                    0B       ARG BUNDLE_TYPE CDS JAVA_R
<missing>                                                    0B       ARG BUNDLE_TYPE CDS JAVA_R
<missing>                                                    0B       ARG BUNDLE_TYPE JAVA_RELEA
<missing>                                                    0B       ARG JAVA_RELEASE
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Saved 500+ MB

But the build is not clean now :(

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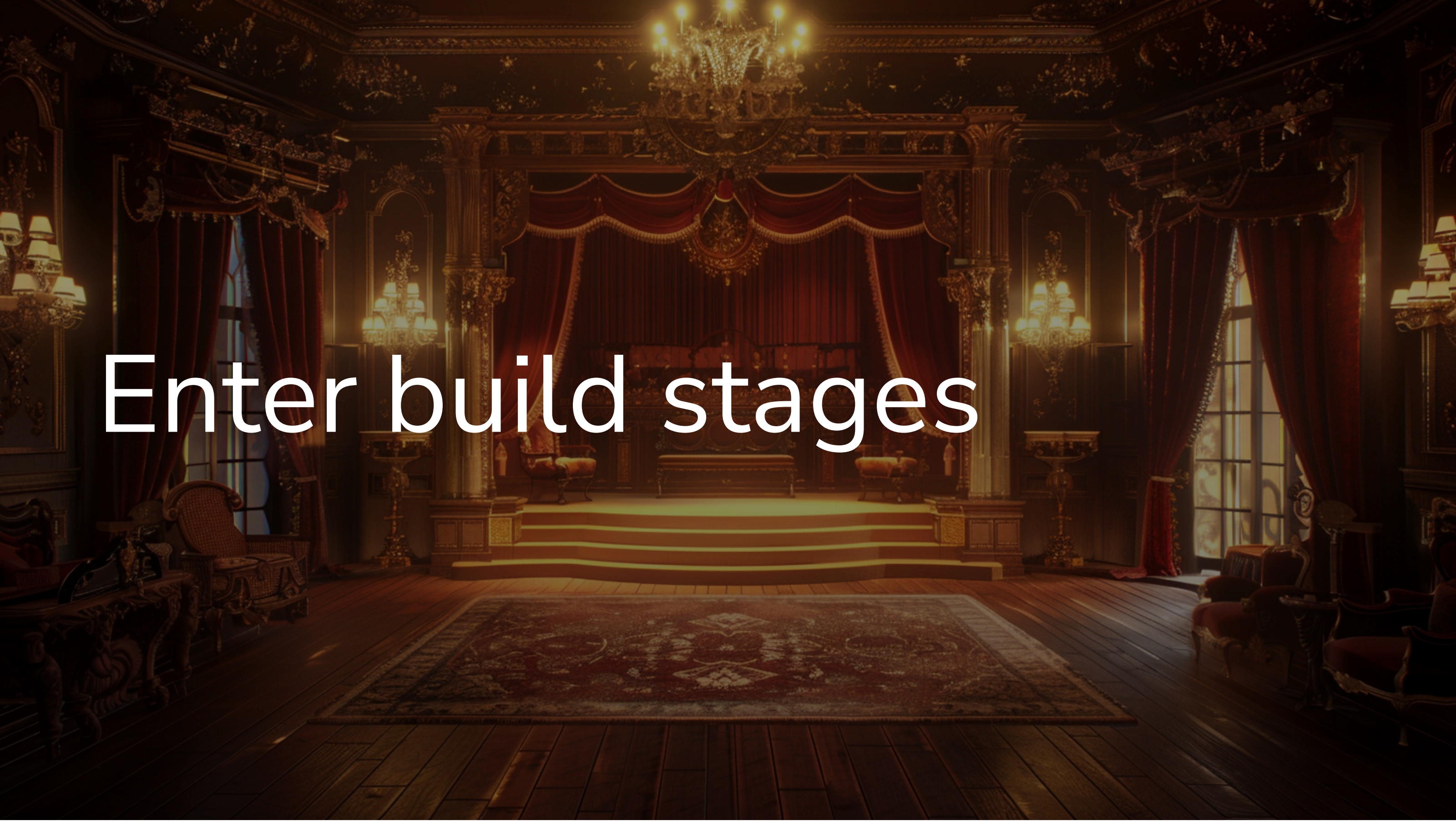
But the build is not clean now :(

We have to build everything outside, what if environment affects the build?



Enter build stages

Multi-stage builds

Holy grail of pure builds

- Allow clean builds
- Allow optimal packaging
- Allow different base images

Staged example

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build -xtest

FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
```

Staged example

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Staged example

```dockerfile
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```

Result

```
ID         TAG                                                SIZE     COMMAND
6ea7ad11af smarter:latest                                     0B       CMD ["/bin/sh" "-c
d6eb71a5df                                                    58.80MiB COPY file:8d96bee9
c6ede8dac2 bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl 98.32MiB |15 BUNDLE_TYPE=jd
<missing>                                                     0B       ENV JAVA_HOME="/us
<missing>                                                     0B       ENV LANG=en_US.UTF
<missing>                                                     0B       LABEL org.opencont
<missing>                                                     0B       LABEL org.opencont
<missing>                                                     0B       LABEL maintainer='
<missing>                                                     0B       ARG BUNDLE_TYPE CD
<missing>                                                     0B       ARG BUNDLE_TYPE CD
<missing>                                                     0B       ARG BUNDLE_TYPE CD
<missing>                                                     0B       ARG BUNDLE_TYPE CD
<missing>                                                     0B       ARG BUNDLE_TYPE JA
<missing>                                                     0B       ARG JAVA_RELEASE
<missing>                                                     7.49MiB  CMD ["/bin/sh"]
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Result

```
ID         TAG                                              SIZE     COMMAND
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c6ede8dac2 bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl 98.32MiB |15 BUNDLE_TYPE=jc
<missing>                                                   0B       ENV JAVA_HOME="/us
<missing>                                                   0B       ENV LANG=en_US.UTF
<missing>                                                   0B       LABEL org.opencont
<missing>                                                   0B       LABEL org.opencont
<missing>                                                   0B       LABEL maintainer='
<missing>                                                   0B       ARG BUNDLE_TYPE CD
<missing>                                                   0B       ARG BUNDLE_TYPE CD
<missing>                                                   0B       ARG BUNDLE_TYPE CD
<missing>                                                   0B       ARG BUNDLE_TYPE CD
<missing>                                                   0B       ARG BUNDLE_TYPE JA
<missing>                                                   0B       ARG JAVA_RELEASE
<missing>                                                   7.49MiB  CMD ["/bin/sh"]
```

That’s all folks!

That’s all folks!

Or is it?

What’s these 59 MiB?

We have to pull 59 MiB of petclinic on every small commit!

Is there a way to optimize it?

Layers!

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build -xtest

FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher
```

- Build image

Layers!

```dockerfile
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

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RUN cd /app && ./gradlew build -xtest

FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher
```

- Build image
- Introduce new "optimizer" stage

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Layers!

```
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WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
```

- Build image
- Introduce new "optimizer" stage

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Layers!

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WORKDIR /app
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FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure

Layers!

```
COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure

Layers!

```
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
- Copy layers

Layers!

```
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
- Copy layers

Layers

Because Spring Boot jar is complex!

```
Usage:
  java -Djarmode=tools -jar my-app.jar

Available commands:
  extract      Extract the contents from the jar
  list-layers  List layers from the jar that can be extracted
```

Layers

```
app
├── application
│   ├── BOOT-INF
│   │   ├── classes
│   │   │   ├── application-mysql.properties
│   │   │   ├── application-postgres.properties
│   │   │   ├── application.properties
│   │   │   ├── banner.txt
│   │   │   ├── db
│   │   │   │   ├── ...
│   │   │   ├── org
```

Layers

```
app
├── application
│   ├── BOOT-INF
│   │   ├── classes
│   │   │   ├── application-mysql.properties
│   │   │   ├── application-postgres.properties
│   │   │   ├── application.properties
│   │   │   ├── banner.txt
│   │   │   ├── db
│   │   │   │   ├── ...
│   │   │   ├── org
```

Layers

```
├── application
│   ├── BOOT-INF
│   │   ├── classes
│   │   │   ├── application-mysql.properties
│   │   │   ├── application-postgres.properties
│   │   │   ├── application.properties
│   │   │   ├── banner.txt
│   │   │   ├── db
│   │   │   │   ├── ...
│   │   │   ├── org
│   │   │   │   └── springframework
```

Layers

```
│   │   │   ├── db
│   │   │   │   ├── ...
│   │   │   ├── org
│   │   │   │   └── springframework
│   │   │   │       ├── aop
│   │   │   │       │   └── aspectj
│   │   │   │       │       └── annotation
│   │   │   │       │           └── AnnotationAwareAspectJAutoProxyCreator__BeanDefinitions.
│   │   │   │       ├── com/github/asm0dey/...
│   │   │   ├── static
│   │   │   ....
```

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Layers

```
│   │   │   │   └── springframework
│   │   │   │       ├── aop
│   │   │   │       │   └── aspectj
│   │   │   │       │       └── annotation
│   │   │   │       │           └── AnnotationAwareAspectJAutoProxyCreator__BeanDefinitions.
│   │   │   │       ├── com/github/asm0dey/ ...
│   │   │   ├── static
│   │   │   ....
│   │   ├── classpath.idx
│   │   └── layers.idx
│   └── META-INF
```

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Layers

```
│   │   │   │       │           └── AnnotationAwareAspectJAutoProxyCreator__BeanDefinitions.
│   │   │   │       ├── com/github/asm0dey/ ...
│   │   │   ├── static
│   │   │   ....
│   │   ├── classpath.idx
│   │   └── layers.idx
│   └── META-INF
│       ├── MANIFEST.MF
│       ├── native-image
│       │   ├── ch.qos.logback
│       │   │   └── logback-classic
```

Layers

```
│   │   │   │       ├── com/github/asm0dey/...
│   │   │   ├── static
│   │   │   ....
│   │   ├── classpath.idx
│   │   └── layers.idx
│   └── META-INF
│       ├── MANIFEST.MF
│       ├── native-image
│       │   ├── ch.qos.logback
│       │   │   └── logback-classic
│       │   │       └── 1.5.6
```

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Layers

```
    └── META-INF
        ├── MANIFEST.MF
        ├── native-image
        │   ├── ch.qos.logback
        │   │   └── logback-classic
        │   │       └── 1.5.6
        │   │           ├── reflect-config.json
        │   │           └── resource-config.json
        │   ├── ...
        ├── sbom
        │   └── application.cdx.json
```

Layers

```
│       │   │       └── 1.5.6
│       │   │           ├── reflect-config.json
│       │   │           └── resource-config.json
│       │   ├── ...
│       ├── sbom
│       │   └── application.cdx.json
│       └── services
│           └── java.nio.file.spi.FileSystemProvider
├── dependencies
│   └── BOOT-INF
│       └── lib
```

Layers

```
│       │   ├── ...
│       ├── sbom
│       │   └── application.cdx.json
│       └── services
│           └── java.nio.file.spi.FileSystemProvider
├── dependencies
│   └── BOOT-INF
│       └── lib
│           ├── angus-activation-2.0.2.jar
│           ├── ...
├── snapshot-dependencies
```

Layers

```
│           └── services
│               └── java.nio.file.spi.FileSystemProvider
├── dependencies
│   └── BOOT-INF
│       └── lib
│           ├── angus-activation-2.0.2.jar
│           ├── ...
├── snapshot-dependencies
└── spring-boot-loader
    └── org
        └── springframework
```

Layers

```
├── dependencies
│   └── BOOT-INF
│       └── lib
│           ├── angus-activation-2.0.2.jar
│           ├── ...
├── snapshot-dependencies
└── spring-boot-loader
    └── org
        └── springframework
            └── boot
                └── loader
```

Layers

```
│   └── BOOT-INF
│       └── lib
│           ├── angus-activation-2.0.2.jar
│           ├── ...
├── snapshot-dependencies
└── spring-boot-loader
    └── org
        └── springframework
            └── boot
                └── loader
                    ├── jar
```

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Layers

```
                    ├── ZipContent$Loader.class
                    ├── ZipContent$Source.class
                    ├── ZipContent.class
                    ├── ZipDataDescriptorRecord.class
                    ├── ZipEndOfCentralDirectoryRecord$Located.class
                    ├── ZipEndOfCentralDirectoryRecord.class
                    ├── ZipLocalFileHeaderRecord.class
                    ├── ZipString$CompareType.class
                    └── ZipString.class

212 directories, 525 files
```

Layered image structure

```
ID         TAG                                               SIZE      COMMAND
618743f6a2 layers:latest                                     2.96MiB   COPY dir:e0faa63b9654
9cca3273f0                                                   0B        COPY dir:acd0d0ac1f7d
fe123ee16e                                                   382.56kiB COPY dir:01225d3c4ef6
dede9bac3d                                                   57.69MiB  COPY dir:755815d928fd
65e3fb4cbf                                                   0B        ENTRYPOINT ["java" "o
7130fa5864 bellsoft/liberica-runtime-container:jre-slim-musl 126.80MiB |18 CDS=no DESCRIPTIO
<missing>                                                    0B        ENV JAVA_HOME="/usr/l
<missing>                                                    0B        ENV LANG=en_US.UTF-8 L
<missing>                                                    0B        LABEL org.opencontain
```

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Layered image structure

```
ID         TAG                                          SIZE      COMMAND
618743f6a2 layers:latest                                2.96MiB   COPY dir:e0faa63b9654
9cca3273f0                                              0B        COPY dir:acd0d0ac1f7d
fe123ee16e                                              382.56kiB COPY dir:01225d3c4ef6
dede9bac3d                                              57.69MiB  COPY dir:755815d928fd
65e3fb4cbf                                              0B        ENTRYPOINT ["java" "o
7130fa5864 bellsoft/liberica-runtime-container:jre-slim-musl 126.80MiB |18 CDS=no DESCRIPTIO
<missing>                                               0B        ENV JAVA_HOME="/usr/l
<missing>                                               0B        ENV LANG=en_US.UTF-8 L
<missing>                                               0B        LABEL org.opencontain
```

- Dependencies: ~57.7MiB

Layered image structure

```
ID         TAG                                             SIZE      COMMAND
618743f6a2 layers:latest                                   2.96MiB   COPY dir:e0faa63b9654
9cca3273f0                                                 0B        COPY dir:acd0d0ac1f7d
fe123ee16e                                                 382.56kiB COPY dir:01225d3c4ef6
dede9bac3d                                                 57.69MiB  COPY dir:755815d928fd
65e3fb4cbf                                                 0B        ENTRYPOINT ["java" "o
7130fa5864 bellsoft/liberica-runtime-container:jre-slim-musl 126.80MiB |18 CDS=no DESCRIPTIO
<missing>                                                  0B        ENV JAVA_HOME="/usr/l
<missing>                                                  0B        ENV LANG=en_US.UTF-8 
<missing>                                                  0B        LABEL org.opencontain
```

- Dependencies: ~57.7MiB
- Launcher: 382.56kiB

Layered image structure

```
ID          TAG                                                SIZE      COMMAND
618743f6a2  layers:latest                                      2.96MiB   COPY dir:e0faa63b96544
9cca3273f0                                                     0B        COPY dir:acd0d0ac1f7df
fe123ee16e                                                     382.56kiB COPY dir:01225d3c4ef6c
dede9bac3d                                                     57.69MiB  COPY dir:755815d928fd9
65e3fb4cbf                                                     0B        ENTRYPOINT ["java" "or
7130fa5864  bellsoft/liberica-runtime-container:jre-slim-musl  126.80MiB |18 CDS=no DESCRIPTION
<missing>                                                      0B        ENV JAVA_HOME="/usr/li
<missing>                                                      0B        ENV LANG=en_US.UTF-8 L
<missing>                                                      0B        LABEL org.opencontaine
```

- Dependencies: ~57.7MiB
- Launcher: 382.56kiB

Layered image structure

```
ID         TAG                                                SIZE      COMMAND
618743f6a2 layers:latest                                      2.96MiB   COPY dir:e0faa63b9654
9cca3273f0                                                    0B        COPY dir:acd0d0ac1f7d
fe123ee16e                                                    382.56kiB COPY dir:01225d3c4ef6
dede9bac3d                                                    57.69MiB  COPY dir:755815d928fd
65e3fb4cbf                                                    0B        ENTRYPOINT ["java" "o
7130fa5864 bellsoft/liberica-runtime-container:jre-slim-musl 126.80MiB |18 CDS=no DESCRIPTIO
<missing>                                                     0B        ENV JAVA_HOME="/usr/l
<missing>                                                     0B        ENV LANG=en_US.UTF-8 L
<missing>                                                     0B        LABEL org.opencontain
```

- Dependencies: ~57.7MiB
- Launcher: 382.56kiB
- Application: ~3MiB

Layered image structure

```
ID         TAG                                              SIZE      COMMAND
618743f6a2 layers:latest                                    2.96MiB   COPY dir:e0faa63b9654
9cca3273f0                                                  0B        COPY dir:acd0d0ac1f7c
fe123ee16e                                                  382.56kiB COPY dir:01225d3c4ef6
dede9bac3d                                                  57.69MiB  COPY dir:755815d928fc
65e3fb4cbf                                                  0B        ENTRYPOINT ["java" "c
7130fa5864 bellsoft/liberica-runtime-container:jre-slim-musl 126.80MiB |18 CDS=no DESCRIPTIO
<missing>                                                   0B        ENV JAVA_HOME="/usr/l
<missing>                                                   0B        ENV LANG=en_US.UTF-8
<missing>                                                   0B        LABEL org.opencontair
```

- Dependencies: ~57.7MiB
- Launcher: 382.56kiB
- Application: ~3MiB

Together: ~61MiB

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61.0MiB > 58.8Mib!

😱

What are we optimizing?

Pull size!

Pull size here is usually only around 3MiB!

We just reinvented how the BellSoft's buildpack works!

And it is amazing

Diversion: buildpacks

https://paketo.io/

Trivial usage:

```
/usr/sbin/pack build petclinic \
  --builder bellsoft/buildpacks.builder \
  --path . \
  -e BP_JVM_VERSION=21
```

Diversion: buildpacks

https://paketo.io/

Trivial usage:

```
/usr/sbin/pack build petclinic \
  --builder bellsoft/buildpacks.builder \
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Diversion: buildpacks

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Diversion: buildpacks

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Trivial usage:

```
/usr/sbin/pack build petclinic \
  --builder bellsoft/buildpacks.builder \
  --path . \
  -e BP_JVM_VERSION=21
```

Result

```
ID         TAG              SIZE      COMMAND
2774e3f214 petclinic:latest 0B        Buildpacks Process Types
<missing>                   1.44kiB   Buildpacks Launcher Config
<missing>                   2.44MiB   Buildpacks Application Launcher
<missing>                   59.17MiB  Application Layer
<missing>                   729.21kiB Software Bill-of-Materials
<missing>                   97.87MiB  Layer: 'jre', Created by buildpack: bellsoft/buildpack
<missing>                   200.00B   Layer: 'java-security-properties', Created by buildpac
<missing>                   4.27MiB   Layer: 'helper', Created by buildpack: bellsoft/buildp
<missing>                   2.97kiB
<missing>                   7.48MiB
```

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Result

```
ID         TAG              SIZE      COMMAND
2774e3f214 petclinic:latest 0B        Buildpacks Process Types
<missing>                   1.44kiB   Buildpacks Launcher Config
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<missing>                   59.17MiB  Application Layer
<missing>                   729.21kiB Software Bill-of-Materials
<missing>                   97.87MiB  Layer: 'jre', Created by buildpack: bellsoft/buildpack
<missing>                   200.00B   Layer: 'java-security-properties', Created by buildpa
<missing>                   4.27MiB   Layer: 'helper', Created by buildpack: bellsoft/build
<missing>                   2.97kiB
<missing>                   7.48MiB
```

Result

```
ID         TAG              SIZE      COMMAND
2774e3f214 petclinic:latest 0B        Buildpacks Process Types
<missing>                   1.44kiB   Buildpacks Launcher Config
<missing>                   2.44MiB   Buildpacks Application Launcher
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<missing>                   97.87MiB  Layer: 'jre', Created by buildpack: bellsoft/buildpack
<missing>                   200.00B   Layer: 'java-security-properties', Created by buildpac
<missing>                   4.27MiB   Layer: 'helper', Created by buildpack: bellsoft/buildp
<missing>                   2.97kiB
<missing>                   7.48MiB
```

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Result

```
ID         TAG              SIZE      COMMAND
2774e3f214 petclinic:latest 0B        Buildpacks Process Types
<missing>                   1.44kiB   Buildpacks Launcher Config
<missing>                   2.44MiB   Buildpacks Application Launcher
<missing>                   59.17MiB  Application Layer
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<missing>                   97.87MiB  Layer: 'jre', Created by buildpack: bellsoft/buildpack
<missing>                   200.00B   Layer: 'java-security-properties', Created by buildpac
<missing>                   4.27MiB   Layer: 'helper', Created by buildpack: bellsoft/buildp
<missing>                   2.97kiB
<missing>                   7.48MiB
```

Buildpacks

Why do we need them?

> - Buildpacks transform your application source code into container images
> - The Paketo open source project provides production-ready buildpacks for the most popular languages and frameworks
> - Use Paketo Buildpacks to easily build your apps and keep them updated

- Reminds of s2i images
- Build better than default
- Spring-aware

Is this all?

We've optimized pull/push size, right?

Optimization is multidimensional

When startup time is more important...

There are several solutions for the Spring application startup time

1. Class Data Sharing (CDS)
2. Coordinated Restore at Checkpoint
3. Native Image

Class Data Sharing (CDS)

- **When:** Java 5 (!)
- **Why:** reduce the startup and memory footprint of multiple JVM instances running on the same host
- **How:** stores data in an archive

How does it help us?

Class Data Sharing (CDS)

- **When**: Java 5 (!)
- **Why**: reduce the startup and memory footprint of multiple JVM instances running on the same host
- **How**: stores data in an archive

How does it help us?

It does not! 🧌

Class Data Sharing (CDS)

- **When**: Java 5 (!)
- **Why**: reduce the startup and memory footprint of multiple JVM instances running on the same host
- **How**: stores data in an archive

How does it help us?

It does not! 🧌But

Application Class Data Sharing

- **When**: Java 10
- **Why**: add application classes to the archive

And then:

- **When**: Java 13, JEP 350
- **Why**: allow addition of classes into the archive upon app exit!

And this helps! How?

How to use AppCDS with Spring?

- `-XX:ArchiveClassesAtExit=application.jsa` to create an archive
- `-Dspring.context.exit=onRefresh` to start and immediately exit the application

NB:

1. Use the same JDK
2. Use the same arguments

And even better!

Spring AOT

Add `-Dspring.aot.enabled=true`

Even more classes!!!

Practice

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build -xtest

FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
```

Practice

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build -xtest

FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
```

Practice

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build -xtest

FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
```

Practice

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build -xtest

FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher

FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
```

Practice

```
#omitted
FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
RUN java -Dspring.aot.enabled=true \
  -XX:ArchiveClassesAtExit=./application.jsa \
  -Dspring.context.exit=onRefresh \
  org.springframework.boot.loader.launch.JarLauncher
```

Practice

```
#omitted
FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner

ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies/ ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
RUN java -Dspring.aot.enabled=true \
  -XX:ArchiveClassesAtExit=./application.jsa \
  -Dspring.context.exit=onRefresh \
  org.springframework.boot.loader.launch.JarLauncher
```

Practice

```
#omitted
FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner

ENTRYPOINT ["java", \
            "-Dspring.aot.enabled=true", \
            "-XX:SharedArchiveFile=application.jsa", \
            "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies / ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
RUN java -Dspring.aot.enabled=true \
  -XX:ArchiveClassesAtExit=./application.jsa \
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```

Practice

```
#omitted
FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner

ENTRYPOINT ["java", \
            "-Dspring.aot.enabled=true", \
            "-XX:SharedArchiveFile=application.jsa", \
            "org.springframework.boot.loader.launch.JarLauncher"]
COPY --from=optimizer /app/app/dependencies / ./
COPY --from=optimizer /app/app/spring-boot-loader/ ./
COPY --from=optimizer /app/app/snapshot-dependencies/ ./
COPY --from=optimizer /app/app/application/ ./
RUN java -Dspring.aot.enabled=true \
  -XX:ArchiveClassesAtExit=./application.jsa \
  -Dspring.context.exit=onRefresh \
  org.springframework.boot.loader.launch.JarLauncher
```

What does it cost ?

```
-r--r--r--          0:0       81 MB  ├── application.jsa
```

Which is not small at all!

What does it cost ?

```
-r--r--r--          0:0       81 MB  ├── application.jsa
```

Which is not small at all!

But you're trading pull speed for startup speed

How much faster?

Up to 50-60%

Pushing further with CRaC

CRaC: Coordinated Restore at Checkpoint

> The CRaC (Coordinated Restore at Checkpoint) Project researches coordination of Java programs with mechanisms to checkpoint (make an image of, snapshot) a Java instance while it is executing. Restoring from the image could be a solution to some of the problems with the start-up and warm-up times. The primary aim of the Project is to develop a new standard mechanism-agnostic API to notify Java programs about the checkpoint and restore events. Other research activities will include, but will not be limited to, integration with existing checkpoint/restore mechanisms and development of new ones, changes to JVM and JDK to make images smaller and ensure they are correct.

https://openjdk.org/projects/crac/

In a perfect world it should be:

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build

FROM bellsoft/liberica-runtime-container:jre-crac-slim as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/ap

FROM bellsoft/liberica-runtime-container:jre-crac-slim as runner

ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
COPY --from=optimizer /app/app.jar /app/app.jar
COPY --from=optimizer /checkpoint /checkpoint
```

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In a perfect world it should be:

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
RUN cd /app && ./gradlew build

FROM bellsoft/liberica-runtime-container:jre-crac-slim as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/ap

FROM bellsoft/liberica-runtime-container:jre-crac-slim as runner

ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
COPY --from=optimizer /app/app.jar /app/app.jar
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```

But in reality

```
#12 6.051       Suppressed: java.lang.RuntimeException: Native checkpoint failed.
#12 6.051               at java.base/jdk.crac.Core.translateJVMExceptions(Core.java:114) ~[
#12 6.051               at java.base/jdk.crac.Core.checkpointRestore1(Core.java:192) ~[na:na
#12 6.051               at java.base/jdk.crac.Core.checkpointRestore(Core.java:299) ~[na:na
#12 6.051               at java.base/jdk.crac.Core.checkpointRestore(Core.java:278) ~[na:na
#12 6.051               at java.base/javax.crac.Core.checkpointRestore(Core.java:73) ~[na:na
#12 6.051               at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(
#12 6.051               at java.base/java.lang.reflect.Method.invoke(Method.java:580) ~[na:
#12 6.051               at org.crac.Core$Compat.checkpointRestore(Core.java:141) ~[crac-1.4
#12 6.051               ... 17 common frames omitted
```

Pasha Finkelshteyn @asm0dey.site @asm0di0

CRaC is hard!

Let’s try to fix it with arcane magic

```
FROM bellsoft/liberica-runtime-container:jdk-musl as builder

COPY . /app
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COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN --security=insecure java -Dspring.context.checkpoint=onRefresh \
  -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar

FROM bellsoft/liberica-runtime-container:jre-crac-slim as runner

ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
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Pasha

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```dockerfile
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FROM bellsoft/liberica-runtime-container:jre-crac-slim as optimizer

COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
WORKDIR /app
RUN --security=insecure java -Dspring.context.checkpoint=onRefresh \
  -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar || true

FROM bellsoft/liberica-runtime-container:jre-crac-slim as runner

ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
```

Pasha

And this is not all!

Did you hear about `buildx`?

```
docker buildx create --buildkitd-flags \
    '--allow-insecure-entitlement security.insecure' \
    --name insecure-builder
```

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```

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Did you hear about `buildx`?

```
docker buildx use insecure-builder
```

And this is not all!

Did you hear about `buildx` ?

```
docker buildx build \
    --allow security.insecure \
    -f Dockerfile.crac \
    -t pet-crac --output type=docker .
```

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Did you hear about `buildx`?

```
docker run --rm -it --privileged pet-crac
```

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```
Restarting Spring-managed lifecycle beans after JVM restore
HikariPool-1 - Thread starvation or clock leap detected (housekeeper delta=1d19h37m42s102ms798µs806ns)
Tomcat started on port 8080 (http) with context path '/'
Restored PetClinicApplication in 0.186 seconds (process running for 0.19)
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- If "It Works" is enough for you -> YES
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3. Run the container with capabilities:
    1. CAP_SYS_PTRACE
    2. CAP_CHECKPOINT_RESTORE
4. Container will run and stop
5. Commit the container like

    ```
    docker commit container-id new-tag
    ```

Pros and Cons

Pros:

1. Does not require arcane magic
2. Works more predictably
3. Does not depend on unstable features
4. Does not require privileged containers in

Cons:

1. Organization-specific
2. Requires more steps

And now we have all flavours of ultimate docker images!

Quick summary?

1. Use layers for faster deployment
2. Use CDS for faster startup without many compromises
3. Use CRaC for a *lightning-fast* startup (with compromises)
4. Use native image if you’re prepared to sacrifice some build time

Thank you!

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END